

Air Quality Prediction

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Project Objectives

- Problem Statement
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project
- Result/outcomes
- Conclusion
- Future Perspective



Problem Statement

- Air pollution is a major environmental and public health concern.
- Poor air quality leads to respiratory and cardiovascular diseases.
- Manual monitoring of pollution is time-consuming and expensive.
- There is a need for an intelligent system that can predict pollution levels using historical sensor data.
- AI can help governments and citizens take preventive actions.

Project Overview

EcoAir AI is a Machine Learning-based system that predicts Carbon Monoxide (CO) concentration using environmental sensor data.

Key Features

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Machine Learning prediction
- Feature importance analysis
- Air pollution prediction
- Supports environmental sustainability

Algorithm Used

- Random Forest Regression

End Users

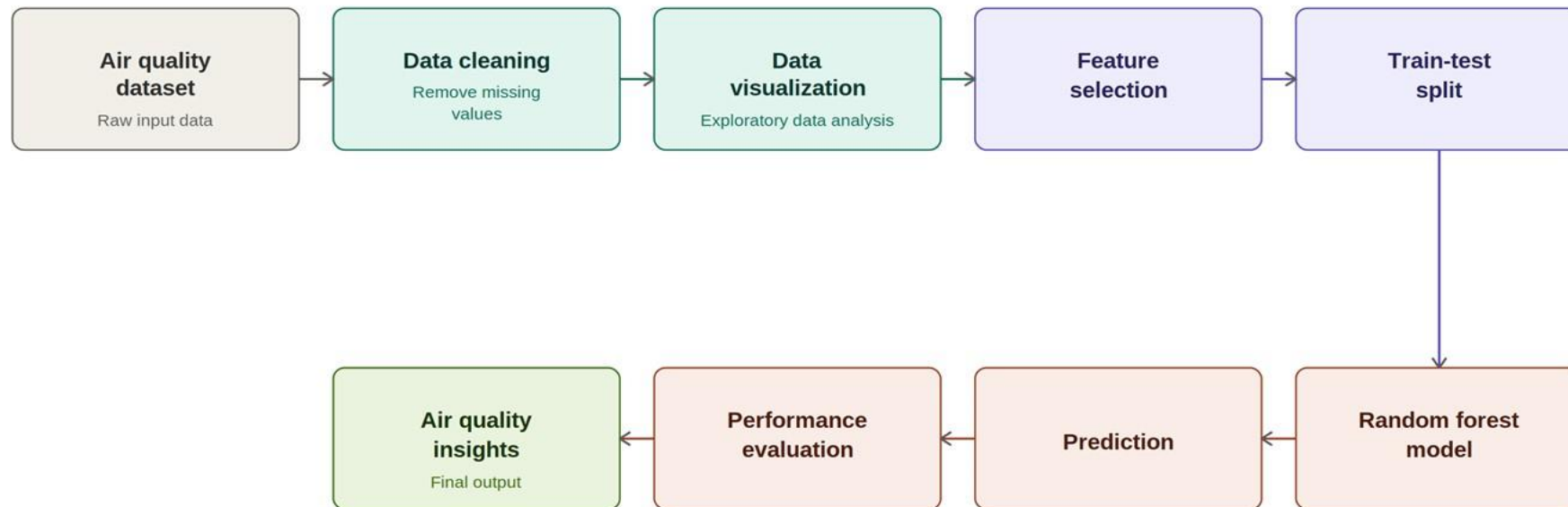
- Environmental Protection Agencies
- Government Pollution Control Boards
- Smart City Authorities
- Research Organizations
- Industries
- Students & Researchers
- General Public

Wow Factor in Project

- Uses Artificial Intelligence for environmental protection
- Predicts pollution levels before critical conditions occur
- Visualizes pollution trends using interactive graphs
- Identifies the most influential pollution factors
- Helps support Sustainable Development Goals (SDG 11 & SDG 13)

Project Flow / Block Diagram

Air Quality Prediction Pipeline



Technologies Used

Programming

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset

- UCI Air Quality Dataset

Machine Learning

- Random Forest Regression

Results / Outcomes

Model Performance

- R^2 Score $\approx$ **0.64**
- Successfully predicts Carbon Monoxide (CO) concentration
- Identifies major environmental factors affecting air quality

Visualizations

- Correlation Heatmap
- Distribution Plot
- Feature Importance

Conclusion

- AI can effectively analyze environmental data.
- Machine Learning helps predict pollution levels.
- Early prediction supports better environmental planning.
- The project demonstrates how AI contributes to Green Sustainability.
- The model can be integrated into smart city monitoring systems.

Future Perspective

- Real-time IoT sensor integration
- Live AQI prediction dashboard
- Mobile application
- Deep Learning models
- Weather API integration
- Smart city deployment
- Pollution alerts and notifications
- Multi-city pollution forecasting

Thank You !!

Github link :- <https://github.com/mymachine0001-ir/Edunet-GTU.git>

Video (google drive link) :- https://drive.google.com/drive/folders/1SwAKNn-JM_pTv5K2A8eQaTLR_wzNZQsQ?usp=sharing